

A plain-language glossary for crystal symmetry in ISDF–THC (diamond as the running example)

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This is a short dictionary of the symmetry words we have been using — *star*, *little group*, *high symmetry*, *orbit*, *copy*, and so on. Each entry is one plain sentence, a quick picture, and a diamond number where it helps. Read it top to bottom: the terms build on each other. Diamond has $|P| = 48$ point-group operations and $n_{\text{IP}} = 136$ interpolation points per cell.

1. The crystal and its symmetries

Unit cell / lattice. The unit cell is the smallest repeating box of atoms; copying it on a regular grid (the lattice) builds the whole crystal.

Space group. The complete list of moves that leave the crystal looking identical — rotations, reflections, and translations, and combinations of them. Diamond’s is called $Fd\bar{3}m$.

Point group. The moves that hold one point fixed (rotations and reflections, no sliding). Diamond has $|P| = 48$ of them. Think of it as the symmetry of a single site.

Symmorphic vs. nonsymmorphic. Symmorphic means every symmetry is a clean rotation/reflection (plus a whole-cell shift). Nonsymmorphic means some symmetries need a *fractional* shift — a “glide” or “screw.” Diamond is nonsymmorphic. In practice this only sprinkles in extra phases; none of the ideas below change.

2. Momentum space

k -point (momentum). A label for how a wave’s phase steps from one cell to the next. Rather than follow every cell, we follow a momentum k . There are N_k of them.

BvK supercell. Born–von Kármán: a finite box of N_k cells with wrap-around (periodic) edges. $N_k = \text{number of cells} = \text{number of } k\text{-points}$ (8 for a $2 \times 2 \times 2$ mesh, 216 for $6 \times 6 \times 6$).

Brillouin zone. The space of allowed k ’s. A few special spots have names — Γ (the center), X , L .

3. Grouping the k -points (the heart of it)

Star (a.k.a. family). Take one k and hit it with all 48 point-group moves. You get a *set* of k ’s — that is its star. Every k in a star is a symmetry-copy of the others, so you only need to study *one* of them. A star has at most 48 members.

Little group (of k). The subset of the 48 moves that leave *this particular k* where it is (up to a whole reciprocal-lattice jump). Rule of thumb (orbit–stabilizer):

$$(\text{star size}) \times (\text{little-group size}) = 48.$$

A large little group means few star members; a small little group means many.

High-symmetry vs. generic (low-symmetry) k . A *high-symmetry k* has a *big* little group (lots of moves fix it) and therefore a *small* star. A *generic / low-symmetry k* has a *small* little group and a *big* star (up to 48). Γ is the most symmetric point — its little group is all 48 moves. Coarse k -meshes are made almost entirely of high-symmetry points; as you refine the mesh, the new points are mostly low-symmetry.

4. The interpolation grid

Interpolation points (ISDF), n_{IP} . ISDF represents products of orbitals by their values at a small, cleverly chosen set of real-space points — the interpolation points. Diamond uses $n_{\text{IP}} = 136$ per cell.

Fiber. The bundle of n_{IP} interpolation points sitting “over” one k . The whole grid is N_k fibers $\times n_{\text{IP}}$ points = $N_k \cdot n_{\text{IP}}$ in total.

Orbit (of interpolation points). Fix one k and apply its little-group moves to one interpolation point: you sweep out a *set* of points — an orbit. Points only ever mix *within* an orbit, never across orbits. An orbit is at most as big as the little group. (Do not confuse this with a star, which is an orbit of k -points; here it is an orbit of *interpolation points*.)

5. Representation-theory words

Representation. A rule that turns each symmetry move into a matrix acting on some space — here, how the moves shuffle-and-phase the interpolation points.

Irreducible representation (irrep). A representation that cannot be split into smaller independent pieces — the “atoms” of symmetry. Each irrep has a fixed size, its *dimension*.

Dimension and partner (d_μ, ν). An irrep of dimension d_μ comes with d_μ *partners* ($\nu = 1, \dots, d_\mu$) that the symmetry moves rotate into one another — like the (x, y) pair under rotation. Diamond’s little-group irreps have dimensions 1, 2, or 3.

Multiplicity and copy (n_μ, c). How many separate times a given irrep appears in a space; each appearance is a *copy*, labelled c . Symmetry cannot tell the copies apart — so this is exactly where the real, material-specific numbers live.

Character. The trace (sum of the diagonal) of an irrep’s matrices. A basis-independent *fingerprint* that identifies the irrep.

Character projection. A formula built from characters that automatically extracts the part of a space belonging to a chosen irrep. It is the non-commutative cousin of a Fourier filter.

Schur’s lemma / commutant. The theorem behind all the savings: any operator that respects the symmetry must be *identity on the partners* and *free on the copies*, and block-diagonal by irrep. That is precisely why symmetry compresses the data.

6. Putting it together: the transform and the blocks

Q (symmetry adaptation). The change of basis from the raw labels $|k\rangle|I\rangle$ (momentum, interpolation point) to the symmetry labels $|i\rangle|a\rangle|c\rangle$. It is the “generalized Fourier transform” of the whole space group (ordinary Fourier on the translations, character projection on the point group).

The symmetry labels (i, a, c) .

- $i = (f, \mu)$: which *star* f and which little-group *irrep* μ — together, the full-crystal irrep.
- $a = (\kappa, \nu)$: which *member* κ of the star, and which *partner* ν of the irrep — together, the partner index (size $m_i = |\text{star}| \cdot d_\mu$).
- c : the *copy* — the multiplicity slot, where the material data sits.

W block structure. In the symmetry basis the Coulomb tensor is $W = \bigoplus_i \mathbb{K}_{m_i} \otimes w_i$: *block-diagonal by $i = (f, \mu)$, identity on the partner $a = (\kappa, \nu)$* (so all star members and all partners share one block), and a *free matrix* w_i only on the copies c .

Fold-back lattice vector (L) . When a symmetry move carries an interpolation point out of its home cell into a neighbour, L records which neighbour. It supplies the Bloch phase $e^{-ik \cdot L}$ that relates one star member to another.

Unique data. The count of genuinely independent numbers left after symmetry: $\sum_{\text{orbits}} |o|^2$ for the transform (6800 at $2 \times 2 \times 2$), $\sum_i n_i^2$ for W (3384). The remarkable part: these stay almost *flat* as N_k grows (the added low-symmetry points contribute tiny blocks), even though N_k itself grows fast.